Topic 10 – Electricity and circuits

Students should:		Maths skills
10.1	Describe the structure of the atom, limited to the position, mass and charge of protons, neutrons and electrons	5b
10.2	Draw and use electric circuit diagrams representing them with the conventions of positive and negative terminals, and the symbols that represent cells, including batteries, switches, voltmeters, ammeters, resistors, variable resistors, lamps, motors, diodes, thermistors, LDRs and LEDs	5b
10.3	Describe the differences between series and parallel circuits	
10.4	Recall that a voltmeter is connected in parallel with a component to measure the potential difference (voltage), in volt, across it	
10.5	Explain that potential difference (voltage) is the energy transferred per unit charge passed and hence that the volt is a joule per coulomb	1a, 1c 3c
10.6	Recall and use the equation: energy transferred (joule, J) = charge moved (coulomb, C) × potential difference (volt, V) $E = Q \times V$	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d
10.7	Recall that an ammeter is connected in series with a component to measure the current, in amp, in the component	
10.8	Explain that an electric current as the rate of flow of charge and the current in metals is a flow of electrons	
10.9	Recall and use the equation: charge (coulomb, C) = current (ampere, A) × time (second, s) $Q = I \times t$	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d
10.10	Describe that when a closed circuit includes a source of potential difference there will be a current in the circuit	
10.11	Recall that current is conserved at a junction in a circuit	
10.12	Explain how changing the resistance in a circuit changes the current and how this can be achieved using a variable resistor	
10.13	Recall and use the equation: potential difference (volt, V) = current (ampere, A) × resistance (ohm, Ω) $V = I \times R$	1a, 1d 2a 3a, 3c, 3d
10.14	Explain why, if two resistors are in series, the net resistance is increased, whereas with two in parallel the net resistance is decreased	

Students should:		Maths skills
10.15	Calculate the currents, potential differences and resistances in series circuits	1a, 1d 2a 3a, 3c, 3d
10.16	Explain the design and construction of series circuits for testing and measuring	
10.17	*Core Practical: Construct electrical circuits to:* *a investigate the relationship between potential difference, current and resistance for a resistor and a filament lamp* *b test series and parallel circuits using resistors and filament lamps*	1a, 1c, 1d 2a, 2b, 2f 3a, 3b, 3c, 3d 4a, 4b, 4c, 4d, 4e
10.18	Explain how current varies with potential difference for the following devices and how this relates to resistance a filament lamps b diodes c fixed resistors	2g 4a, 4b, 4c, 4d, 4e
10.19	Describe how the resistance of a light-dependent resistor (LDR) varies with light intensity	4c, 4d
10.20	Describe how the resistance of a thermistor varies with change of temperature (negative temperature coefficient thermistors only)	4c, 4d
10.21	Explain how the design and use of circuits can be used to explore the variation of resistance in the following devices a filament lamps b diodes c thermistors d LDRs	5b
10.22	Recall that, when there is an electric current in a resistor, there is an energy transfer which heats the resistor	
10.23	Explain that electrical energy is dissipated as thermal energy in the surroundings when an electrical current does work against electrical resistance	
10.24	Explain the energy transfer (in 10.22 above) as the result of collisions between electrons and the ions in the lattice	
10.25	**Explain ways of reducing unwanted energy transfer through low resistance wires**	
10.26	Describe the advantages and disadvantages of the heating effect of an electric current	

Students should:		Maths skills
10.27	Use the equation: energy transferred (joule, J) = current (ampere, A) × potential difference (volt, V) × time (second, s) $E = I \times V \times t$	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d
10.28	Describe power as the energy transferred per second and recall that it is measured in watt	1c
10.29	Recall and use the equation: power (watt, W) = energy transferred (joule, J) ÷ time taken (second, s) $P = \frac{E}{t}$	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d
10.30	Explain how the power transfer in any circuit device is related to the potential difference across it and the current in it	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
10.31	Recall and use the equations: electrical power (watt, W) = current (ampere, A) × potential difference (volt, V) $P = I \times V$ electrical power (watt, W) = current squared (ampere^2, A^2) × resistance (ohm, Ω) $P = I^2 \times R$	1a, 1b, 1c, 1d 2a 3a, 3b, 3c, 3d
10.32	Describe how, in different domestic devices, energy is transferred from batteries and the a.c. mains to the energy of motors and heating devices	
10.33	Explain the difference between direct and alternating voltage	4c
10.34	Describe direct current (d.c.) as movement of charge in one direction only and recall that cells and batteries supply direct current (d.c.)	
10.35	Describe that in alternating current (a.c.) the movement of charge changes direction	
10.36	Recall that in the UK the domestic supply is a.c., at a frequency of 50 Hz and a voltage of about 230 V	
10.37	Explain the difference in function between the live and the neutral mains input wires	
10.38	Explain the function of an earth wire and of fuses or circuit breakers in ensuring safety	
10.39	Explain why switches and fuses should be connected in the live wire of a domestic circuit	

Students should:		Maths skills
10.40	Recall the potential differences between the live, neutral and earth mains wires	
10.41	Explain the dangers of providing any connection between the live wire and earth	
10.42	Describe, with examples, the relationship between the power ratings for domestic electrical appliances and the changes in stored energy when they are in use	1c 2c

Use of mathematics

- Make calculations using ratios and proportional reasoning to convert units and to compute rates (1c, 3c).
- Apply the equations relating p.d., current, quantity of charge, resistance, power, energy, and time, and solve problems for circuits which include resistors in series, using the concept of equivalent resistance (1c, 3b, 3c, 3d).
- Use graphs to explore whether circuit elements are linear or non-linear and relate the curves produced to their function and properties (4c, 4d).
- Make calculations of the energy changes associated with changes in a system, recalling or selecting the relevant equations for mechanical, electrical, and thermal processes; thereby express in quantitative form and on a common scale the overall redistribution of energy in the system (1a, 1c, 3c).

Suggested practicals

- Investigate the power consumption of low-voltage electrical items.